Notes: Berry and Haile (Econometrica, 2014)

Identification in Differentiated Products Markets Using Market Level Data

Contents

1 Big picture

- **Research question.** What can be identified in differentiated-products demand and supply models when the researcher observes only market-level data, such as prices, product characteristics, market shares, and instruments, but not individual choices or firm marginal costs?
- **Main idea.** The paper asks whether the identifying content of BLP-style models comes from parametric functional forms and distributional assumptions, or from more basic economic structure plus instrumental-variable variation.
- **Demand side.** The demand model is a nonparametric random utility discrete-choice model with rich preference heterogeneity, product-market demand shocks, and endogenous prices.
- **Supply side.** The supply model allows nonparametric cost functions and latent cost shocks and nests standard oligopoly models, including marginal-cost pricing, monopoly or joint-profit maximization, Bertrand pricing, and Cournot quantity setting.
- **Core finding.** Demand, welfare changes, marginal costs, cost functions, and some restrictions on firm conduct can be identified nonparametrically, provided the researcher has sufficiently rich instruments and the model satisfies key monotonicity and invertibility conditions.
- **Key practical message.** "BLP instruments" based on rival product characteristics are useful and often necessary, but in the fully nonparametric market-level setting they are generally not sufficient by themselves. Additional excluded shifters, such as cost shifters or proxies for costs, are typically needed.
- **Economic takeaway.** The standard parametric assumptions used in applied IO are not the fundamental source of identification. Their main roles are finite-sample approximation, extrapolation, and compensation for limited exogenous variation.

2 Incremental contribution of the paper

- **Nonparametric foundation for BLP-type market data models.** The paper provides general identification results for differentiated-products models with market-level data, endogenous prices, product-market unobservables, and flexible consumer heterogeneity.
- **Demand identification with product-market unobservables.** Prior discrete-choice identification results typically lacked the combination of rich heterogeneous preferences and endogenous prices induced by product-market demand shocks. Berry and Haile address that combination directly.
- **Clarifies the role of instruments.** The paper explains why rival characteristics can help identify demand even though they enter the demand system, and why additional excluded instruments are needed in the fully nonparametric case.
- **Links demand and supply.** Once demand is identified, standard oligopoly first-order conditions identify markups and marginal costs. With additional exclusion restrictions, the marginal cost functions themselves can be identified.
- **Two complementary identification strategies.** The first strategy uses nonparametric IV completeness conditions. The second treats demand and supply as a system of nonparametric simultaneous equations and gives constructive proofs using excluded demand and cost shifters.

- **Formalizes Bresnahan's conduct intuition.** The paper generalizes the idea that market-level variation can discriminate between oligopoly models. The results allow differentiated products, heterogeneous firms, latent demand and cost shocks, and supply models beyond conjectural variations.
- **Guidance for applied work.** The paper makes clear the tradeoff between functional form assumptions and exclusion restrictions. More flexible demand requires richer exogenous variation.

3 Model and key objects

3.1 Market-level observables

A market t contains a continuum of consumers facing a choice set $\mathcal{J} = \{0, 1, \ldots, J\}$, where good 0 is the outside good. The observed market-level variables include:

- market size M_t;
- product characteristics $x_t = (x_{1t}, \ldots, x_{Jt})$;
- prices $p_t = (p_{1t}, \ldots, p_{Jt})$;
- market shares $s_t = (s_{1t}, \ldots, s_{Jt})$;
- demand instruments z_t, excluded from consumer utility;
- later, cost shifters w_t and cost-side instruments y_t.

Interpretation. This is the common empirical IO setting: the researcher sees products, prices, shares, and instruments across markets, but not individual consumer choices or firms' realized marginal costs.

3.2 Random utility demand

Consumer i in market t has utilities

$$(v_{i0t}, v_{i1t}, \ldots, v_{iJt}), \qquad v_{i0t} = 0.$$

Conditional on market characteristics, utilities are iid across consumers and markets. Market shares are choice probabilities:

$$s_{jt} = \sigma_j(\chi_t) = \Pr\left(\arg\max_{k \in \mathcal{J}} v_{ikt} = j \mid \chi_t\right), \qquad j = 0, 1, \ldots, J,$$

where $\chi_t = (x_t, p_t, \xi_t)$ and $\xi_t = (\xi_{1t}, \ldots, \xi_{Jt})$ are product-market demand unobservables.

Interpretation. The shocks ξ_{jt} are known to firms and consumers but unobserved by the econometrician. Because firms set prices while knowing ξ_{jt}, prices are endogenous.

3.3 Index restriction

The paper partitions observed characteristics into $x_{jt} = (x_{jt}^{(1)}, x_{jt}^{(2)})$ and imposes an index restriction:

$$\delta_{jt} = x_{jt}^{(1)}\beta_j + \xi_{jt}.$$

After normalizing the scale, the paper writes

$$\delta_{jt} = x_{jt}^{(1)} + \xi_{jt}.$$

The distribution of utilities may depend on ξ_t and $x_t^{(1)}$ only through the indices δ_t, while dependence on $x_t^{(2)}$ and p_t remains flexible.

- This is weaker than a linear random coefficients model in many dimensions, because the utility distribution can be otherwise nonparametric.
- It is still a substantive restriction: $x_{jt}^{(1)}$ and ξ_{jt} must be perfect substitutes in the demand index.
- Appendix B shows that the linear index can be relaxed to a monotone nonseparable index under stronger IV assumptions.

Interpretation. The index restriction is what makes exogenous product characteristics usable as instruments in the inverted demand equations.

3.4 Connected substitutes

The connected-substitutes condition requires that goods behave like substitutes in both $-p_t$ and δ_t.

- Raising the attractiveness index δ_{jt} of good j weakly reduces other goods' shares.
- Lowering the price of good j weakly reduces other goods' shares.
- There is enough strict substitution across the choice set to avoid disconnected blocks of goods.

Interpretation. This condition is central because it allows the market-share system to be inverted. It is a nonparametric analog of the Berry (1994) inversion for logit-style demand.

3.5 Supply side

On the supply side, the paper lets $q_{jt} = M_t s_{jt}$ be quantity and lets mc_{jt} denote the equilibrium marginal cost of product j in market t. The model is high level: supply is summarized by first-order conditions that can be solved for marginal costs:

$$mc_{jt} = \psi_j\big(s_t, M_t, D(\chi_t), p_t\big),$$

where $D(\chi_t)$ is the matrix of price derivatives of demand.

- Under a known conduct model, ψ_j is known.
- Under a weaker version, ψ_j exists but is not known.

- The paper shows that standard static oligopoly models imply this form under connected substitutes and differentiability.

Interpretation. The supply model is flexible enough to nest many conduct assumptions while retaining the key implication that demand and observed prices reveal implied marginal costs.

4 Methodology and identification results

4.1 Demand inversion

Under the index and connected-substitutes assumptions, market shares can be inverted. For any observed (s_t, p_t), there is at most one vector δ_t that rationalizes those shares at those prices:

$$\delta_{jt} = \sigma_j^{-1}(s_t, p_t), \qquad j = 1, \ldots, J.$$

Using $\delta_{jt} = x_{jt} + \xi_{jt}$, the inverted demand equation is

$$x_{jt} + \xi_{jt} = \sigma_j^{-1}(s_t, p_t),$$

or equivalently,

$$\xi_{jt} = \sigma_j^{-1}(s_t, p_t) - x_{jt}.$$

Interpretation. This equation has the flavor of a nonparametric IV regression, but with a nonstandard feature: all endogenous variables (s_t, p_t) enter inside the unknown inverse-demand function.

4.2 Nonparametric IV identification of demand

The first identification strategy assumes:

$$E[\xi_{jt} \mid z_t, x_t] = 0,$$

and a completeness condition: if

$$E[B(s_t, p_t) \mid z_t, x_t] = 0,$$

then $B(s_t, p_t) = 0$ almost surely.

Theorem 1 states that, under the demand assumptions and these IV conditions:

- each demand shock ξ_{jt} is identified with probability one;
- each demand function $\sigma_j(\chi_t)$ is identified on its support.

Interpretation. The proof follows the Newey–Powell logic for nonparametric IV models. The inverse-demand function is the only function satisfying the conditional moment restriction.

4.3 Why both x_t and excluded instruments matter

In a fully nonparametric model, inverse demand depends on the $2J$ endogenous variables (s_t, p_t). The vector x_t provides J instruments because it enters the inverted equation with a known normalized coefficient. The remaining variation must come from excluded instruments such as cost shifters, markup shifters, or other variables that affect prices and shares without entering utility.

Interpretation. This is why rival characteristics are useful but generally not enough. Product characteristics help instrument for market shares, while cost or markup shifters help instrument for prices.

4.4 Consumer welfare

For welfare, the paper adds quasilinear preferences:

$$v_{ijt} = \mu_{ijt} - p_{jt},$$

with μ_{it} independent of p_t conditional on (x_t, ξ_t). Under this restriction, aggregate consumer surplus changes from a price vector p^0 to p^1 can be written as a line integral of demand:

$$\int_a^b \sigma(\delta_t, \alpha(\tau)) \cdot \alpha'(\tau)\, d\tau,$$

where $\alpha(a) = p^0$ and $\alpha(b) = p^1$.

- Price-change welfare effects are identified on path-connected subsets of the support of $p_t \mid \delta_t$.
- Welfare effects of characteristic changes require stronger support conditions, because they require integrating demand over larger price regions.
- Figure 1 illustrates how price movements reveal probability mass in choice regions.

Interpretation. Market-level data generally cannot identify individual welfare changes, but with quasilinearity they can identify aggregate welfare changes when the necessary price support is available.

4.5 Marginal costs

Once demand is identified, the matrix of demand derivatives $D(\chi_t)$ is identified. If the supply model is known and implies

$$mc_{jt} = \psi_j\big(s_t, M_t, D(\chi_t), p_t\big),$$

then marginal costs are identified. This is Theorem 2.

- No additional cost-side exclusion restriction is needed to recover equilibrium marginal costs.
- The result applies to the standard supply models listed in the paper, including marginal cost pricing, monopoly pricing, Bertrand pricing, and Cournot quantity-setting.
- The recovery of marginal costs is a nonparametric version of the standard BLP supply-side markup inversion.

Interpretation. If the researcher is willing to specify conduct, prices and demand slopes reveal implied markups and marginal costs.

4.6 Marginal cost functions

To identify the marginal cost function, the paper adds a cost equation such as

$$mc_{jt} = c_j(Q_{jt}, w_{jt}) + \omega_{jt},$$

where Q_{jt} is the vector of quantities produced by the firm that makes product j, w_{jt} are observed cost shifters, and ω_{jt} is a cost shock.

With cost-side instruments y_{jt} satisfying

$$E[\omega_{jt} \mid w_{jt}, y_{jt}] = 0$$

and a corresponding completeness condition, Theorem 3 identifies:

- the marginal cost function $c_j(Q_{jt}, w_{jt})$;
- the cost shock ω_{jt}.

Interpretation. Once marginal costs have been recovered, cost-function identification becomes a standard nonparametric IV problem.

4.7 Cost shocks without specifying conduct

The paper also studies how to identify cost shocks without committing to a known supply model. It assumes a cost index

$$\kappa_{jt} = w_{jt} + \omega_{jt},$$

and a monotone marginal cost function in κ_{jt}. Then the supply-side equilibrium relation can be inverted:

$$w_{jt} + \omega_{jt} = \pi_j^{-1}(s_t, p_t).$$

With appropriate cost and demand shifters, Theorem 4 identifies ω_{jt} even when the conduct function ψ_j is unknown.

Interpretation. This result is important for conduct testing because it lets the researcher recover cost-side shocks without first assuming which oligopoly model is true.

4.8 Simultaneous equations approach

The second main approach treats demand and supply as a system:

$$x_{jt} + \xi_{jt} = \sigma_j^{-1}(s_t, p_t),$$
$$w_{jt} + \omega_{jt} = \pi_j^{-1}(s_t, p_t).$$

This approach uses excluded demand and cost shifters to trace out the inverse demand and inverse supply relations.

- Theorem 5 gives identification of demand under a large-support condition on (x_t, w_t).
- Theorem 6 gives identification of cost shocks and the reduced-form equilibrium pricing function.

- Theorem 7 relaxes large support by imposing shape restrictions on the density of structural errors; a multivariate Normal is one example satisfying the relevant nonsingularity condition.

Interpretation. This approach replaces abstract completeness assumptions with more constructive variation in demand and cost shifters, but it requires additional structure, including an equilibrium selection condition.

4.9 Discriminating between oligopoly models

The paper then asks whether market-level data can test alternative models of firm conduct. The key idea is that a correct conduct model implies a conduct-specific function ψ_j that maps observed prices, shares, and demand derivatives into marginal cost. If two observations have the same relevant cost arguments, the implied marginal cost must be the same.

In the strongest case, Theorem 8 gives the testable restriction:

$$\psi_j\big(s_t, \widetilde{D}(s_t, p_t), p_t\big) = \psi_{j'}\big(s_{t'}, \widetilde{D}(s_{t'}, p_{t'}), p_{t'}\big)$$

whenever the same cost function and the same (Q, κ) apply across the two observations.

A weaker case, Theorem 9, uses conditional expectation restrictions when the cost index is not identified without specifying conduct.

Interpretation. The paper formalizes the idea that variation in demand and cost shifters can rule out false conduct models. The variation need not literally rotate demand in the old homogeneous-goods sense.

5 Identification and interpretation

5.1 What the paper can identify

The paper establishes identification of several objects:

- the structural demand shocks ξ_{jt};
- the market share functions $\sigma_j(\chi_t)$;
- aggregate consumer welfare changes under quasilinearity and sufficient support;
- equilibrium marginal costs under a known supply model;
- marginal cost functions under additional cost-side IV conditions;
- cost shocks under a cost-index structure;
- testable restrictions that can discriminate among conduct models.

Interpretation. This is not an estimation paper. It tells the applied researcher what information is in market-level data under different combinations of model restrictions and instruments.

5.2 Why the paper needs market-level instruments

The central problem is that prices and market shares are endogenous. Prices are endogenous because firms observe product-market demand shocks when setting prices. Shares are endogenous in the inverted demand equation because each share depends on all demand shocks.

The paper's nonparametric inverse demand equation is

$$\xi_{jt} = \sigma_j^{-1}(s_t, p_t) - x_{jt}.$$

Since σ_j^{-1} is completely unknown and depends on all of (s_t, p_t), identification requires instruments that move the entire endogenous vector in rich enough ways.

5.3 Role of BLP instruments

In applied BLP work, characteristics of competing products often serve as instruments. Berry and Haile clarify their logic.

- Own and rival product characteristics help because they shift market shares through the choice set.
- Rival characteristics can identify substitution patterns by changing the attractiveness of alternatives.
- In the fully nonparametric market-level model, these characteristics generally must be supplemented by excluded instruments that move prices, such as cost shifters.

Interpretation. "BLP instruments" are not ad hoc. They are useful because demand inversion turns market shares into endogenous regressors and product characteristics become excluded from some inverse-demand components through the index structure.

5.4 Role of the index restriction

The index restriction does two jobs.

- It turns the demand unobservable into a scalar index for each product-market observation.
- It makes x_{jt} enter the inverse-demand equation with a known normalized coefficient, so x_t can act as part of the instrument set.

Without the index restriction, the rival-characteristic logic would be much weaker. The paper therefore shows that identification is nonparametric but not assumption-free.

5.5 Completeness versus support conditions

The paper offers two families of assumptions.

- The nonparametric IV approach uses completeness conditions, which are the nonparametric analog of rank conditions.

- The simultaneous equations approach uses excluded demand and cost shifters with large support, or smaller support plus restrictions on the density of unobservables.

Interpretation. These are different mathematical ways to say that the data must contain enough independent variation to distinguish one nonparametric function from another.

5.6 Welfare support limitation

Demand identification on observed support does not automatically identify all counterfactual welfare effects. Price changes are identified when the path between old and new prices lies within the support of prices conditional on demand indices. Characteristic changes require stronger support because they require learning demand over regions that may not be observed.

Interpretation. The paper is careful about what can and cannot be learned. Identification of demand on support is not the same as global counterfactual power.

5.7 Conduct testing interpretation

The conduct tests use a simple economic contradiction: if a candidate conduct model implies different marginal costs for observations that should have the same cost arguments, the candidate model is false.

- Figure 2 illustrates the logic with marginal-cost pricing versus monopoly pricing.
- False models imply the wrong mapping from demand curves to marginal costs.
- Cross-market variation in demand and cost shifters can reveal this contradiction.

5.8 Limitations

The results are identification results, not guarantees of easy estimation. In finite samples, nonparametric estimators may be demanding. The exclusion restrictions and completeness or support assumptions are strong. The index and connected-substitutes assumptions impose economic structure. For welfare and conduct counterfactuals, the support of observed variation matters.

6 Tables and figures

6.1 Figure 1: Choice regions and price changes

Read:

- Figure 1 considers the case $J = 2$ and partitions the utility space into regions in which consumers choose good 0, good 1, or good 2.
- Panel (a) shows the choice regions for a given price vector.
- Panel (b) shows how a price change shifts the boundaries between choice regions.
- The hatched regions represent consumers whose choices switch when prices move.

- The figure supports the welfare discussion: price variation reveals probability mass over regions of the unobserved taste distribution.

Economic message: Identified demand plus price variation can reveal aggregate welfare changes. With limited price support, the data may identify only partial information about the full distribution of money-metric utilities.

6.2 Figure 2: Testing oligopoly conduct

Read:

- Figure 2 illustrates the difference between two supply models, such as marginal-cost pricing and monopoly pricing.
- Panel (a) shows that the same observed equilibrium can imply different marginal costs under different conduct assumptions.
- Panel (b) shows how variation that changes the relevant demand or marginal-revenue object while holding cost arguments fixed can rule out a false conduct model.
- The figure is the visual version of Theorem 8's restriction: the correct ψ_j must imply the same marginal cost whenever the relevant cost function and cost arguments are the same.

Economic message: Market-level data can contain testable implications for firm conduct. The paper generalizes Bresnahan's conduct-testing intuition beyond homogeneous goods and conjectural-variation models.

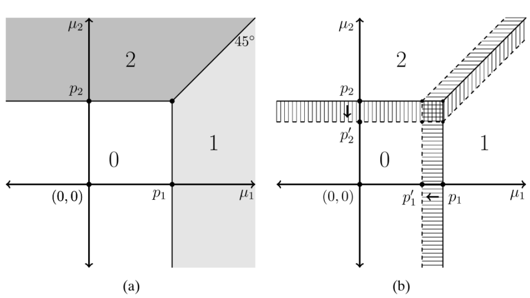

FIGURE 1.—(a) Choice regions with $J = 2$. (b) Price changes move choice regions.

Figure 1: Choice regions and price changes.

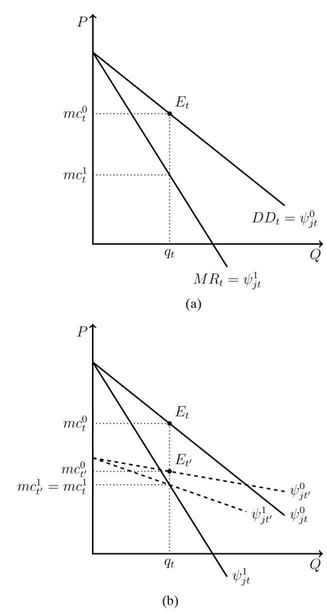

FIGURE 2.—(a) Possible marginal costs under two models. (b) Rotation of ψ_j^1 rules out the false model.

Figure 2: Conduct testing through implied marginal costs.

7 Short summary

- The paper studies identification in BLP-style differentiated-products markets using only market-level data.
- The demand model allows rich preference heterogeneity, product-market demand shocks, and endogenous prices.
- The key demand assumptions are a scalar demand index and connected substitutes.
- These assumptions let the demand system be inverted, producing nonparametric inverse-demand equations.
- Demand is identified under nonparametric IV exclusion and completeness conditions.
- Identification requires both product-characteristic variation and additional excluded instruments that move prices.
- Under quasilinear preferences, aggregate consumer welfare changes from price movements can be identified on the relevant support.
- Given demand, standard supply first-order conditions identify marginal costs.
- With cost-side instruments, marginal cost functions can also be identified.
- A simultaneous-equations approach gives constructive identification results using demand and cost shifters.
- The paper formalizes tests that can discriminate between alternative oligopoly conduct models.
- The central practical lesson is the tradeoff between functional form and instruments.

8 Conclusion

8.1 Core conclusion

Berry and Haile show that differentiated-products demand and supply models can have a strong nonparametric identification foundation even when only market-level data are observed. The essential ingredient is sufficiently rich exogenous variation from instruments, not the specific parametric forms used in most applied work.

8.2 Economic intuition

The intuition is that market shares summarize consumer choices, but observed shares are distorted by unobserved product quality and endogenous prices. If the model lets the researcher invert shares into latent demand indices, instruments can separate true demand from the unobservables. Once demand slopes are known, supply first-order conditions convert prices and shares into markups and marginal costs.

8.3 Contribution revisited

The paper turns common applied IO practice into a set of formal identification results. It shows why the BLP inversion is powerful, why product characteristics can be instruments, why additional cost or markup shifters are needed in flexible models, and why conduct assumptions can sometimes be tested rather than simply imposed.

8.4 Implications

For empirical work, the paper implies that researchers should pay close attention to the source and richness of excluded variation. Parametric assumptions can be useful and sometimes unavoidable in finite samples, but they should be understood as substitutes for unavailable exogenous variation rather than as proof that identification is impossible without them.

8.5 Limitations

The paper is about identification, not implementation. Completeness, support, and exclusion restrictions are strong and can be hard to verify. Nonparametric identification does not guarantee precise nonparametric estimation in realistic data. Counterfactuals that move outside the observed support remain difficult.

8.6 Takeaway

The enduring takeaway is that market-level differentiated-products models are identifiable when economic structure and instruments are rich enough. The real empirical question is therefore not just which functional form to estimate, but whether the data contain the exogenous variation needed to learn demand, costs, welfare, and conduct.